

a.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

$$\bar{x} = 1$$

$$\bar{y} = 2$$

b.

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_1 = \bar{y} - b_2 \bar{x}$$

$$b_2 = \frac{8}{10} = \frac{4}{5}$$

$$b_1 = 2 - \frac{4}{5} \cdot 1 = \frac{6}{5}$$

c.

$$\sum_{i=1}^5 x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^5 x_i y_i = 12 + 4 + 3 + 1 + 0 = 18$$

$$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1 = 10 = \sum (x_i - \bar{x})^2$$

$$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8 = \sum (x_i - \bar{x})(y_i - \bar{y})$$

d.

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	$\frac{18}{5}$	$\frac{2}{5}$	$\frac{4}{25}$	$\frac{6}{5}$
2	2	$\frac{14}{5}$	$-\frac{4}{5}$	$\frac{16}{25}$	$-\frac{8}{5}$
1	3	2	1	1	1
-1	1	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{9}{25}$	$-\frac{3}{5}$
0	0	$\frac{6}{5}$	$-\frac{6}{5}$	$\frac{36}{25}$	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = \frac{18}{5}$	$\sum x_i \hat{e}_i = 0$

$$\hat{y}_i = \frac{6}{5} + \frac{4}{5}x_i$$

$$\hat{e}_i = y_i - \hat{y}_i$$

$$s_y^2 = \frac{4+0+1+1+4}{5-1} = \frac{5}{2}$$

$$s_x^2 = \frac{4+1+0+4+1}{5-1} = \frac{5}{2}$$

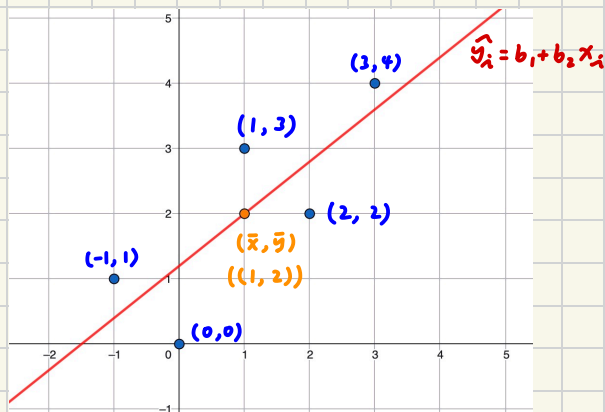
$$s_{xy} = \frac{4+0+0+2+2}{5-1} = 2$$

$$r_{xy} = \frac{2}{\sqrt{\frac{5}{2}} \cdot \sqrt{\frac{5}{2}}} = \frac{4}{5}$$

$$CV_x = 100 \frac{\sqrt{\frac{5}{2}}}{1} = 50\sqrt{10}$$

median of x : 1

e. f.



g. $b_1 = \frac{6}{5}, b_2 = \frac{4}{5}$

Substitute $(1, 2) \Rightarrow 2 = \frac{6}{5} + \frac{4}{5} \cdot 1 \checkmark$

h. $\bar{y} = \frac{\sum \hat{y}_i}{5} = 2 = \bar{y}$

i. $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{5-2}$
 $= \frac{6}{5}$

j. $\widehat{\text{var}}(b_2 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}$
 $\text{se}(b_2) = \sqrt{\widehat{\text{var}}(b_2 | x)}$
 $\widehat{\text{var}}(b_2 | x) = \frac{\frac{6}{5}}{10} = \frac{3}{25}$
 $\text{se}(b_2) = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}$